

A Gated-Fusion Hybrid CNN-Transformer Architecture for Interpretable, Robust, and Cross-Dataset ECG Arrhythmia Detection

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Abstract—Deep learning now matches cardiologist-level accuracy on electrocardiogram (ECG) arrhythmia classification, yet clinical deployment remains limited because most published models are opaque, are evaluated only on clean signals, and lose accuracy when applied outside their training distribution. Prior work on hybrid convolutional neural network (CNN) and Transformer architectures for ECG typically reports a single accuracy figure under one of these conditions and rarely compares more than one explanation method, leaving open how such models behave once interpretability, noise robustness, and cross-dataset transfer are considered together. This paper addresses that gap with a controlled comparison of five CNN and Transformer ablation conditions, including a learnable gated-fusion hybrid, evaluated under one protocol on the MIT-BIH Arrhythmia Database and then examined for explanation quality across three methods, robustness under additive noise, and generalisation to the PTB-XL and INCART databases. The convolutional-only model attains the highest macro F1-score (0.955), while the gated-fusion hybrid offers the most balanced profile once noise robustness and cross-dataset behaviour are also considered. Grad-CAM produces visibly broader relevance regions than Integrated Gradients or Transformer attention on representative examples, and a naive zero-shot transfer to INCART loses over 0.7 of macro F1-score, indicating that domain shift, rather than architecture choice, is the dominant obstacle to deployment. All beats were split by a class-stratified rather than a patient-independent procedure, so the reported figures should be read as an optimistic upper bound pending a record-independent re-evaluation, which is identified as the priority correction for future work.

Index Terms—electrocardiogram, arrhythmia classification, convolutional neural network, Transformer, hybrid architecture, explainable artificial intelligence, domain adaptation, self-supervised learning, MIT-BIH, PTB-XL, INCART

I. INTRODUCTION

Cardiovascular disease remains the leading cause of death worldwide, and the electrocardiogram is still the most widely used, lowest-cost tool for detecting the arrhythmias that precede many cardiac events. Over the past decade, deep learning has repeatedly matched or exceeded cardiologist-level performance on ECG classification benchmarks, moving the field from hand-crafted features toward end-to-end convolutional and, more recently, Transformer-based models [10], [11], [14],

[16]. Despite this progress, three practical barriers continue to separate research prototypes from clinically deployable systems: interpretability, robustness and generalisation.

A clinician cannot be expected to trust a diagnosis produced by a model whose reasoning cannot be inspected, and regulatory frameworks for medical software increasingly require some form of explanation [6], [7]. Ambulatory and wearable recordings are frequently corrupted by baseline wander, muscle artefact and electrode motion, yet most published arrhythmia classifiers are evaluated only on clean, curated data. Finally, a model trained on one population, device and annotation protocol commonly loses a substantial fraction of its accuracy when applied to a different hospital, country or patient cohort, a phenomenon attributed to dataset shift.

Recent studies have begun to address each of these barriers individually: gated hybrid CNN-Transformer architectures for raw accuracy [17], systematic XAI benchmarking for interpretability [23], [24], [25], edge-deployable models for efficiency [29], [30], and cross-domain frameworks for generalisation [26]. What is missing is a single study that evaluates one model family across all of these axes at once, since a model selected purely on clean-signal accuracy is not necessarily the model best suited to a noisy or cross-institutional deployment. This paper reports such a study.

Five ablation conditions of shared architectural components are trained on the MIT-BIH Arrhythmia Database [1] under the AAMI five-superclass scheme: (A) a convolutional-only residual network, (B) a Transformer-only encoder operating on convolutional patch tokens, (C) a sequential hybrid in which convolutional features are tokenised and then contextualised by a Transformer encoder, (D) a parallel dual-branch hybrid with a learnable gate blending convolutional and Transformer features, following the gated-fusion principle recently proposed for ECG classification [17], and (F) a self-supervised pretrained Transformer baseline obtained by masked-reconstruction pretraining followed by supervised fine-tuning, in the spirit of recent ECG foundation models trained at a much larger scale [19], [20], [21]. Each condition

is evaluated on held-out accuracy and macro F1-score, on explanation quality using Grad-CAM [6], Integrated Gradients [7] and Transformer classification-token attention, on robustness to additive white Gaussian noise, and on cross-dataset generalisation through fine-tuning on the twelve-lead PTB-XL dataset [2] and zero-shot evaluation on the INCART database.

This paper makes the following contributions:

- a controlled comparison of convolutional, Transformer and hybrid architectures, including a learnable gated-fusion condition, under one shared training protocol;
- a qualitative evaluation using three explainability methods, Grad-CAM, Integrated Gradients and Transformer attention, on the same trained model;
- a robustness evaluation under additive noise across a wide signal-to-noise ratio range;
- a cross-dataset evaluation combining supervised fine-tuning on PTB-XL and zero-shot testing on INCART, together with an explicit account of the resulting domain-shift gap.

The paper also identifies, and reports transparently rather than concealing, a methodological limitation in the MIT-BIH train-test split that qualifies how the headline numbers should be read; this is discussed in Sections III and V.

The remainder of the paper is organised as follows. Section II surveys related work. Section III describes the datasets, preprocessing, splitting strategy and architectures. Section IV reports results across all four evaluation axes. Section V discusses their implications, and Section VI concludes with limitations and future work.

II. RELATED WORK

A. Convolutional Baselines for ECG Classification

Automated arrhythmia classification using deep convolutional networks was popularised by Rajpurkar et al., who trained a thirty-four-layer convolutional network on single-lead recordings and reported performance exceeding practising cardiologists [10]. Hannun et al. extended this to a broader rhythm taxonomy and a larger training corpus [11]. Acharya et al. showed that a shallow convolutional network on short, fixed-length segments achieves strong performance on MIT-BIH without hand-crafted features [12], while Kachuee et al. demonstrated that representations learned on one arrhythmia dataset transfer to a second through fine-tuning [13]. At the twelve-lead level, Ribeiro et al. trained a residual network on more than two million exams with high sensitivity and specificity [14], and Attia et al. showed a convolutional model can screen for asymptomatic left ventricular dysfunction [15]. Strodtz et al. established PTB-XL as a standardised benchmark and showed residual and Inception-style convolutional architectures remain the strongest single-branch baselines [16], consistent with the convolutional-only condition achieving the highest raw macro F1-score in the present study.

B. Transformers and Hybrid CNN-Transformer Architectures

The Transformer architecture replaces recurrence and convolution with multi-head self-attention and has since been

adapted to sequence and, through patch tokenisation, image-like modalities [4], [5]. Self-attention offers a natural mechanism for modelling long-range dependencies that pure convolutional stacks can only approximate through depth, and because a single arrhythmia class can present both as a local morphological anomaly and a longer-range rhythmic irregularity, several recent studies combine convolutional and Transformer branches rather than treating them as competitors. Xiong et al. introduced HCTG-Net, in which a residual convolutional branch and a Transformer branch are processed in parallel and combined through a learnable, per-dimension gate rather than simple concatenation, reporting 99.46 percent accuracy and F1 of 0.9711 on MIT-BIH [17]. Wahid et al. proposed a slimmer ResNet-Vision-Transformer hybrid for myocardial infarction detection, replacing standard patch embedding with smaller convolutions augmented by channel attention [18]. The gated-fusion mechanism evaluated here as condition D is a direct methodological descendant of Xiong et al.’s architecture, adapted to a single-lead, five-class setting and evaluated jointly with explainability and robustness criteria the original study did not consider.

C. ECG Foundation Models and Self-Supervised Pretraining

A parallel direction moves away from fully supervised training toward self-supervised pretraining on large unlabelled ECG corpora followed by lightweight fine-tuning. McKeen et al. released ECG-FM, a Transformer foundation model pretrained on 1.5 million twelve-lead recordings using masked reconstruction and contrastive multi-segment coding [19]. Narayanswamy et al. pretrained a modified Vision Transformer on more than 800,000 unlabelled recordings and fine-tuned it for twelve clinical prediction tasks, finding pretraining outperformed training from scratch, particularly with limited labels [20]. Li et al. reported the largest ECG foundation model to date, trained on more than ten million recordings [21]. Huynh et al. addressed foundation models’ sensitivity to corrupted or missing leads by jointly training on clean signals, corrupted signals and paired text reports [22]. These systems are trained on datasets three to four orders of magnitude larger than the one used in this study; the self-supervised pretrained Transformer baseline evaluated here as condition F follows the same masked-reconstruction principle at a scale appropriate for a single-GPU project, and is named accordingly rather than as a foundation model, since the openly released ECG-FM checkpoint could not be retrieved at the time of writing.

D. Explainable AI for ECG Deep Learning

Because clinical adoption depends on a physician’s ability to audit model reasoning, a growing literature applies post-hoc explainability to ECG models. Manimaran et al. conducted the most comprehensive systematic review to date, screening 6,448 candidate studies and analysing fifty-one, and found the majority rely on a narrow set of conventional techniques with almost no standardised evaluation of explanation faithfulness [23]. Goettling et al. benchmarked thirteen XAI methods

against a convolutional ECG classifier using perturbation-based fidelity and human-interpretability ratings, averaged over many samples, and found Deep Taylor Decomposition most trustworthy, with Grad-CAM ranking below several gradient-based alternatives [24]. Bender et al. applied Layer-wise Relevance Propagation and Integrated Gradients to a twelve-lead classifier and showed both align closely with cardiology diagnostic criteria [25]. Both studies base their conclusions on aggregate statistics across a full test set, which the present paper does not attempt; the comparison in Section IV is illustrative rather than statistical, a distinction discussed further in Section VI.

E. Cross-Dataset Generalisation and Multimodal Alignment

Models trained and evaluated on a single database routinely overstate real-world performance, since databases differ in sampling rate, lead configuration, demographics and annotation protocol. Li, Aierken, Xu et al. proposed a Transformer-based cross-domain framework combining a domain-invariant feature-enhancement module with an interpretability-driven attention constraint, reporting an 8.4 percent accuracy improvement over a domain-naïve baseline on MIT-BIH and a 4.7 percent F1 improvement on INCART [26]. Ansari et al. surveyed Transformer- and large-language-model-based approaches to ECG diagnosis, observing that cross-modal alignment with clinical text substantially improves zero-shot generalisation [27]. Phan et al. proposed multimodal, multi-lead contrastive pretraining that reduces label dependency and improves cross-dataset robustness [28]. The zero-shot INCART evaluation and batch-normalisation domain-alignment step used before PTB-XL fine-tuning here are a direct, lightweight response to this literature.

F. Lightweight and Edge-Deployable Models

A final strand addresses the computational cost of hybrid and Transformer-based models, a prerequisite for wearable deployment. Busia et al. demonstrated the first full Transformer deployed on a microcontroller, achieving a footprint under ninety kilobytes and nine-millisecond latency through quantisation and pruning while retaining 99.6 percent accuracy on MIT-BIH [30]. Hizem et al. implemented a complementary TinyML pipeline for ECG anomaly detection on Raspberry Pi and Arduino hardware [29]. Huang et al. reviewed the TinyML landscape, introducing micro-trainers, small on-device training loops that personalise a model without centralised data sharing [31]. Parameter counts for all five ablation conditions are reported in Section IV to situate the proposed architectures within this budget context.

G. Summary of Identified Gaps

Across these areas, four gaps recur that no single prior study addresses simultaneously: the absence of a learnable gated-fusion condition compared against sequential hybrid and single-branch baselines under matched training; reliance on a single XAI method without a faithfulness-oriented comparison; near-total absence of noise-robustness evaluation for

hybrid CNN-Transformer ECG models; and omission of an explicit domain-adaptation step across databases. The design in Section III addresses all four within one pipeline, while Section VI is explicit about which of them are fully closed by the present experiments and which remain open.

III. MATERIALS AND METHODS

A. Datasets

Three public ECG resources are used, summarised in Table I. The MIT-BIH Arrhythmia Database [1] provides forty-eight half-hour, two-lead recordings at 360 Hz, annotated beat-by-beat and mapped onto five AAMI superclasses: Normal (N), Supraventricular (S), Ventricular (V), Fusion (F) and Unknown/paced (Q). INCART provides seventy-five half-hour, twelve-lead Holter recordings at 257 Hz and is used exclusively as a cross-domain target, since its population, rate and lead configuration differ from MIT-BIH. PTB-XL [2] provides 21,837 twelve-lead recordings from 18,885 patients labelled with one or more of five diagnostic superclasses, and is used to test transfer under a genuinely different, multi-label task formulation.

TABLE I: Summary of the Three ECG Datasets Used in This Study

Property	MIT-BIH	INCART	PTB-XL
Sampling rate	360 Hz	257 Hz	500 Hz
Leads	2	12	12
Records	48	75	21,837
Units used	109,452 beats	175,811 beats	17,259 records
Label unit	beat	beat	record
Label scheme	AAMI (5-class)	AAMI (5-class)	5 superclasses
Role in study	train/test	zero-shot target	transfer target

B. Signal Preprocessing and Dataset Harmonisation

MIT-BIH and INCART were harmonised on a common beat-level representation despite their different sampling rates and lead layouts. For MIT-BIH, the first recorded channel (the modified limb lead II in nearly all records) is used; for INCART, lead index 1 in the stored signal array, corresponding to lead II, is used, so that both datasets contribute a lead-II-equivalent signal. Each channel is passed through a fourth-order Butterworth bandpass filter, applied with zero-phase forward-backward filtering to avoid group delay, followed by a 50 Hz IIR notch filter for powerline interference. The bandpass cutoffs are not identical between the two databases: MIT-BIH uses 0.5 to 40 Hz, the conventional diagnostic band, while INCART uses a wider 0.5 to 100 Hz band, automatically clipped to below the Nyquist frequency, chosen for the earlier exploratory analysis of the INCART signals; this difference in high-frequency content is a harmonisation detail that a stricter protocol would eliminate by applying identical cutoffs to both databases, and is noted here rather than left implicit.

Rather than resampling either signal to a common sampling rate, the two datasets are harmonised by matching the physical duration of the beat window instead of the sample count. R-peaks are taken from the expert beat annotations, and each

MIT-BIH beat is segmented into a window of ninety samples before and one hundred and sixty samples after the R-peak, a 250-sample, 694-millisecond window at 360 Hz. The INCART window is calibrated to the same 694-millisecond duration at its native 257 Hz rate, giving eighty samples before and ninety-eight after the R-peak, a 178-sample window. Every beat, from either database, is then normalised independently to zero mean and unit variance (per-beat z-score). Finally, the same AAMI mapping table is applied to the raw annotation symbols of both databases, so that a given annotation symbol, for example a premature ventricular contraction, is mapped to the same one of the five AAMI superclasses regardless of which database it came from. Figure 1 illustrates the MIT-BIH filtering step on a ten-second segment of record 100.



Fig. 1: Ten-second segment of MIT-BIH record 100 showing the raw modified-limb-lead-II signal (top), the same signal after bandpass and notch filtering (middle), and the second available raw lead, V5, for reference (bottom).

C. Exploratory Analysis and Class Imbalance

Figure 2 shows the resulting class distribution for MIT-BIH and INCART; both are heavily dominated by the Normal class, with Fusion comprising fewer than one percent of beats in either. Figure 3 shows mean beat morphology per AAMI class after z-score normalisation; the classes are visibly separable at the mean-waveform level, confirmed quantitatively by a t-SNE embedding [8] of two thousand sampled beats (Figure 4), where Normal beats form well-separated clusters while minority classes form smaller, partially overlapping groups. To correct class imbalance without discarding data, SMOTE-Tomek resampling [3] is applied to the training partition only, oversampling minority classes while removing Tomek-link boundary pairs (Figure 5); validation and test partitions are left untouched, and class weights from the pre-resampling distribution are additionally passed to the loss function as a second imbalance-handling mechanism.

D. Data Splitting Strategy and a Methodological Limitation

The MIT-BIH beats used for model training and evaluation are split 70/15/15 into training, validation and test partitions using scikit-learn’s stratified `train_test_split`, stratified on the AAMI class label. This stratification is applied *after* beats from all forty-eight records have been pooled into a single array, and no record or patient identifier is retained at

split time. Consequently, the split is a class-stratified beat-level split, not a patient-independent or record-independent split: two beats from the same recording, and therefore from the same patient, can and very likely do appear on opposite sides of the train/test boundary.

This matters because consecutive or nearby beats from one patient share morphological characteristics, baseline noise and electrode placement that are specific to that recording. A model can partially learn to recognise the recording rather than the arrhythmia, which would inflate held-out accuracy relative to what the model would achieve on an unseen patient. This is a well-documented risk in the ECG classification literature and is precisely the kind of leakage that a reviewer should expect to see ruled out. It was not ruled out in the present study, and the headline results in Section IV should therefore be read as an optimistic, beat-level upper bound on performance rather than as evidence of patient-level generalisation. A patient-independent split, for example grouping beats by source record with `GroupShuffleSplit` or an explicit leave-records-out partition before any stratification, is identified as the single highest-priority correction for future work and is a precondition for treating these results as representative of clinical performance. The PTB-XL transfer split (Section III-G) is a simple random 80/10/10 partition over whole records rather than beats, which does not share this particular risk, since PTB-XL labels are already assigned at the record level.

E. Model Architectures

All five ablation conditions are built from two shared building blocks, shown in Figure 6 for the gated-fusion condition. The convolutional branch is a compact one-dimensional residual network with a strided stem followed by three residual blocks (kernel size seven, increasing channel width), terminating in global average pooling to a 128-dimensional feature; the unpooled final feature map is retained as the Grad-CAM target. The Transformer branch tokenises the input via strided convolutional patch embedding (approximately twenty-five tokens per beat), prepends a learnable CLS token, adds sinusoidal positional encodings, and processes the sequence with a four-layer, four-head encoder using GELU activations; an attention probe tied to the final layer extracts CLS-to-token attention for explainability. Both branches adapt automatically to the input length and channel count of any dataset.

The five conditions combine these blocks as follows. Condition A (CNN-only) pools the convolutional branch into a two-layer MLP head. Condition B (Transformer-only) uses only the CLS-token output. Condition C (sequential hybrid) feeds the unpooled convolutional feature sequence into the Transformer encoder as its token sequence. Condition D (gated-fusion hybrid, Figure 6) runs both branches in parallel and computes a learnable, per-dimension sigmoid gate from their concatenated pooled features, directly implementing the gated-fusion principle of Xiong et al. [17]. Condition E reuses condition D’s trained weights with no new parameters as an explainability layer (Section III-H). Condition F, the self-supervised pre-trained Transformer baseline, pretrains the Transformer branch

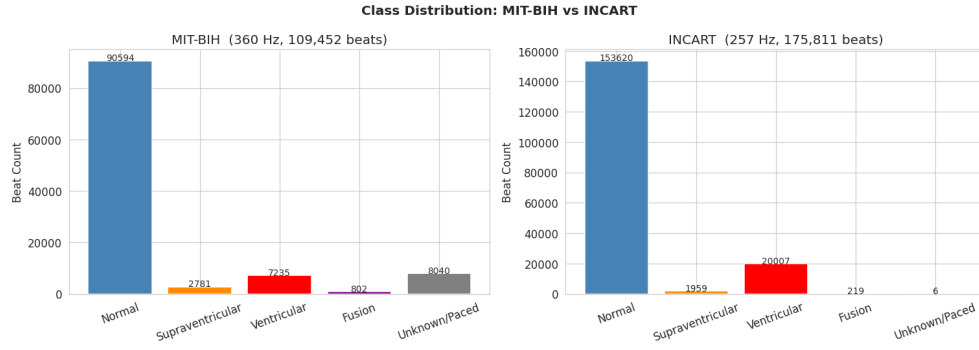


Fig. 2: Beat-level class distribution for the MIT-BIH Arrhythmia Database (left) and the INCART database (right) after AAMI five-class mapping. Both datasets are dominated by the Normal class and both contain very few Fusion beats, motivating the resampling strategy described in Section III-D.

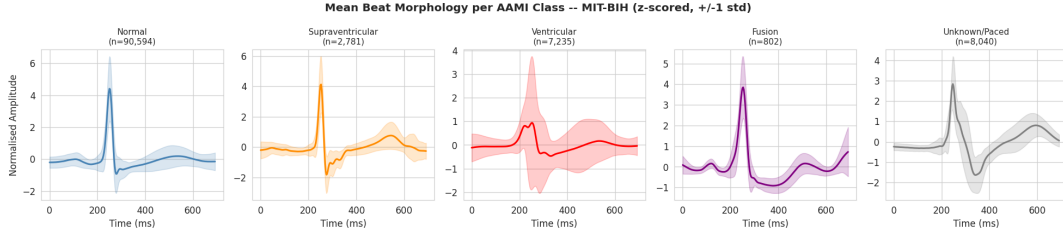


Fig. 3: Mean beat morphology per AAMI class on MIT-BIH after z-score normalisation, with shading indicating one standard deviation. The five classes are visibly distinguishable at the level of the mean waveform even before any model is trained.

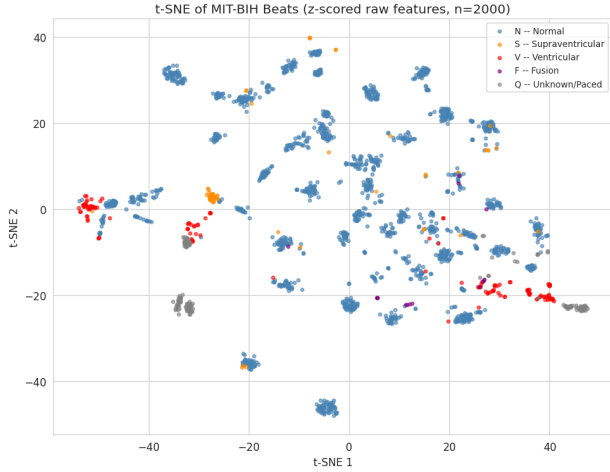


Fig. 4: t-SNE embedding of 2,000 randomly sampled, z-score-normalised MIT-BIH beats. The Normal class (blue) forms the bulk of well-separated clusters, while Ventricular (red), Supraventricular (orange) and Unknown/Paced (grey) beats form smaller, partially overlapping groups.

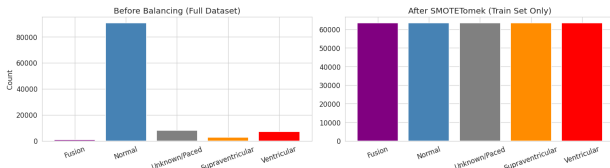


Fig. 5: Training-set class counts before (left) and after (right) SMOTETomek resampling. Resampling is applied to the training partition only; validation and test partitions retain the original, imbalanced distribution.

with a masked-patch reconstruction head for ten epochs, then fine-tunes for up to fifteen; this approximates the pretrain-then-fine-tune paradigm of published ECG foundation models [19], [20], [21] at a scale tractable on a single Colab GPU.

Because the five conditions combine a different number of copies of the two shared building blocks, most obviously condition D contains both branches in full while condition A contains only one, their parameter counts are not matched. Table II reports the exact count for each condition, ranging from 465,445 parameters for the smallest (CNN-only) to 1,367,205 for the largest (gated-fusion hybrid), a near three-fold difference that is treated as part of the experimental record rather than as a controlled variable; comparisons across conditions in Section IV should therefore be read as comparisons between complete, differently sized architectural designs, not as a strict capacity-controlled ablation.

F. Training Protocol

All models are trained with AdamW [9] (initial learning rate 1×10^{-3} , weight decay 1×10^{-4}), gradients clipped to norm 5.0. A plateau scheduler halves the learning rate after two stagnant validation-F1 epochs; training stops early after six stagnant epochs, restoring the best checkpoint. All models train for up to thirty epochs, batch size 128, with class-weighted cross-entropy (MIT-BIH) or class-weighted binary cross-entropy (PTB-XL). Full settings are summarised in Table II. Every condition was trained once, with a fixed random seed (42) controlling weight initialisation, data shuffling and the resampling procedure; no repeated runs with different seeds were performed, because of compute constraints on a shared

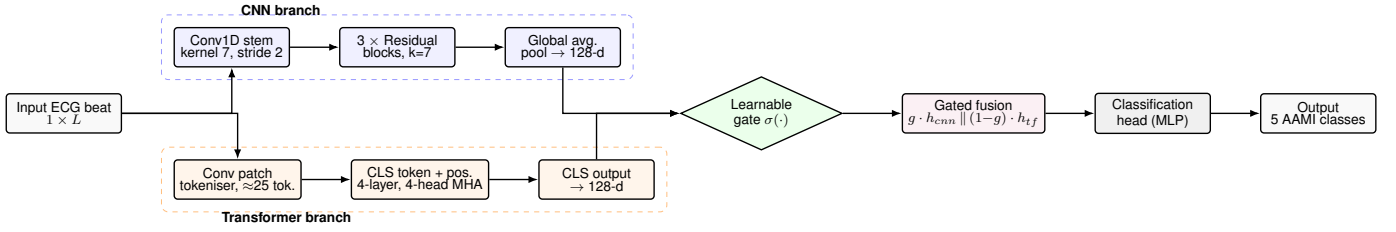


Fig. 6: Architecture of the gated-fusion hybrid (ablation condition D). The CNN branch (top) and Transformer branch (bottom) run in parallel on the same input beat; a learnable, per-dimension sigmoid gate combines their pooled 128-dimensional features before a shared classification head. Conditions A to C and F reuse the same two branches with the gate and, where relevant, the parallel path removed or reordered, as described in the text.

TABLE II: Architecture and Training Settings for Reproducibility

Setting	Value
CNN stem	Conv1d, kernel 7, stride 2
CNN body	3 residual blocks, kernel 7
CNN pooled feature	128-d, global average pool
Transformer tokeniser	strided Conv1d, ≈ 25 tokens
Transformer encoder	4 layers, 4 heads, GELU
Transformer feature	128-d, CLS token
Gate (condition D)	sigmoid, per-dimension
Classification head	2-layer MLP
Params A / B / C / D / F	465K / 869K / 1.33M / 1.37M / 869K
Optimiser	AdamW, lr 1×10^{-3} , wd 1×10^{-4}
LR schedule	plateau, factor 0.5, patience 2
Early stopping	patience 6 (val. macro F1)
Batch size	128
Max epochs	30 (10+15 for condition F)
Loss	class-weighted cross-entropy
Gradient clipping	max norm 5.0
Random seed	42 (single run per condition)

Colab GPU allocation. This is a limitation of the statistical strength of the comparisons in Section IV and is discussed further in Section VI: differences of a few tenths of a macro F1 point between conditions, in particular between the gated-fusion and sequential hybrids, should not be over-interpreted as robust architectural effects without repeated-seed confirmation.

G. Explainability Methods

Three complementary methods are compared on the trained gated-fusion model, following the multi-method evaluation approach of Goettling et al. and Bender et al. [24], [25]. Grad-CAM [6] computes a class-discriminative map from gradients into the CNN branch’s final feature map. Integrated Gradients [7], via the Captum library, accumulates gradients along a path from a zero baseline, giving an axiomatically grounded attribution independent of any single layer. Transformer CLS attention is extracted from the attention probe of Section III-F. Unlike Goettling et al. and Bender et al., who average perturbation-based fidelity metrics over a full test set, the comparison performed here applies all three methods to a single representative beat per AAMI class (Figure 9) and is intentionally described as a qualitative illustration in Section IV rather than as quantitative evidence of relative faithfulness; a full-test-set faithfulness evaluation is identified as required future work in Section VI.

H. Noise Robustness Protocol

Additive white Gaussian noise is injected into the frozen test set at seven SNR levels (-5, 0, 5, 10, 15, 20, 30 dB), following the protocol used in the edge-deployment and signal-robustness literature [22], [29], [30]. Noise power is computed per batch from empirical signal power. Macro F1 is recomputed at every level for conditions A through D (Figure 10).

I. Cross-Dataset Transfer Protocol

Two experiments use the trained gated-fusion model. In the first, the trained MIT-BIH model body is transferred to a freshly instantiated network for the twelve-lead, multi-label PTB-XL task by copying every parameter tensor whose name and shape match between the source and target networks; of 116 tensors in the target network, 114 are copied directly, covering the entire CNN branch, Transformer branch and gating mechanism, while the remaining two, the input stem’s first convolution (one input channel for MIT-BIH versus twelve for PTB-XL) and the final classification layer (five AAMI classes versus five multi-label PTB-XL superclasses with a different output activation), differ in shape and are left randomly initialised. Before fine-tuning, the batch-normalisation layers of the transferred body have their running statistics reset and recomputed with a calibration forward pass over the PTB-XL training split only, using a cumulative moving average (an AdaIN-style realignment), after which standard momentum is restored; this step is motivated by the domain-shift evidence reported by Li, Aierken, Xu et al. [26]. The realigned model is then fine-tuned for fifteen epochs with binary cross-entropy on a random 80/10/10 record-level split. In the second experiment, the MIT-BIH-trained gated-fusion model is applied directly, with no fine-tuning or alignment, to INCART beats, to obtain a zero-shot estimate of the raw domain-shift penalty.

IV. RESULTS

A. Ablation Comparison on MIT-BIH

Table III reports accuracy, macro F1, precision and recall for all five conditions on the held-out test partition, with parameter counts repeated from Table II for convenience; Figure 7 ranks the macro F1-scores. As discussed in Section III-D, the test partition shares its patient pool with the training partition, so these figures are an optimistic, beat-level estimate

rather than a patient-level generalisation estimate, and as discussed in Section III-E each figure reflects a single training run. With that caveat, the convolutional-only model achieves the highest macro F1 (0.9554), consistent with Strodtz et al.’s finding that convolutional architectures remain a strong baseline on curated ECG benchmarks [16]. The gated-fusion and sequential hybrids are close (0.9465 and 0.9460) and both outperform the Transformer-only baseline (0.9263). The self-supervised pretrained Transformer baseline attains 0.9423 and the highest macro recall of any condition (0.9613), suggesting pretraining is effective at reducing false negatives on rare classes even without surpassing the fully supervised hybrids at this dataset size.

TABLE III: Ablation Comparison on the MIT-BIH Test Set

Model	Acc.	F1	Prec.	Rec.	Params
A. CNN-only	0.9926	0.9554	0.9634	0.9478	465,445
D. Gated hybrid	0.9914	0.9465	0.9439	0.9491	1,367,205
C. Sequential hybrid	0.9906	0.9460	0.9508	0.9415	1,326,757
F. Self-sup. Transformer	0.9906	0.9423	0.9262	0.9613	869,253
B. Transformer-only	0.9851	0.9263	0.9034	0.9523	869,253

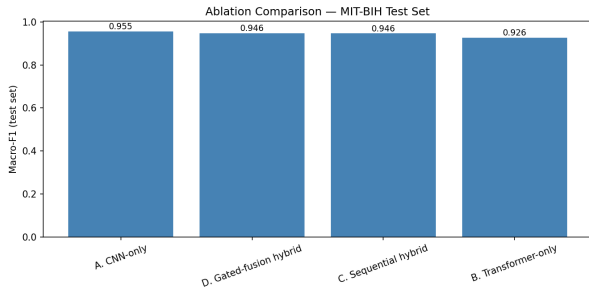


Fig. 7: Macro F1-score of the four fully supervised ablation conditions on the MIT-BIH test set, ranked from highest to lowest.

Figure 8 shows the confusion matrices of conditions A through D, enlarged for readability. All four classify Normal and Unknown/Paced almost perfectly, with residual error concentrated between Normal and the rare Fusion and Supraventricular classes, consistent with the class imbalance in Figure 2 and the morphological overlap in Figure 3.

B. Explainability Comparison

Figure 9 compares Grad-CAM, Integrated Gradients and Transformer CLS attention for one representative beat of each class, all on the gated-fusion model, enlarged from the earlier draft for legibility. Grad-CAM produces broad, contiguous relevance regions often spanning more than half the waveform, for example nearly the entire QRS-to-T-wave region for Fusion and Ventricular. Integrated Gradients produces markedly sparser attributions concentrated around the R-peak and, for several classes, a secondary cluster near the T-wave, visually closer to the clinically relevant markers reported by Bender et al. [25]. Transformer attention is intermediate in granularity, and for Ventricular and Supraventricular isolates a narrow region centred on the abnormal QRS complex.

These observations are consistent with Goettling et al.’s finding that Grad-CAM is not necessarily the most faithful method on one-dimensional ECG data [24], but it must be stressed that the comparison here uses one beat per class rather than the full test set and no quantitative fidelity metric, unlike Goettling et al. and Bender et al., who average such metrics with reported variability over many samples. The pattern shown in Figure 9 is therefore reported as a qualitative illustration consistent with the wider literature, not as independent statistical evidence that Integrated Gradients or attention is more faithful than Grad-CAM; a quantitative, full-test-set faithfulness evaluation with reported uncertainty is required before that stronger claim could be made, and is listed as required future work in Section VI.

C. Noise Robustness

Figure 10 shows macro F1 against additive Gaussian noise SNR for the four supervised conditions. All degrade sharply below ten decibels, falling from above 0.94 on clean data to below 0.35 at zero decibels and below 0.23 at minus five decibels, confirming none possess an intrinsic robustness advantage under severe noise. Between five and fifteen decibels the sequential hybrid is consistently most robust, and the Transformer-only model overtakes the convolutional-only model despite lower clean-signal accuracy, suggesting self-attention confers some resilience to perturbation that pure convolution does not. Above twenty decibels the four curves converge, indicating the differences in Table III are primarily a clean-signal phenomenon. As with Table III, these curves reflect a single training run per condition, and the same seed was used to generate the noise realisation for every model at a given SNR so that comparisons at a fixed noise level are at least free of noise-sampling variation between models, though not free of training-run variation.

D. Cross-Dataset Generalisation

Table IV reports the two cross-dataset experiments. After batch-normalisation alignment and fifteen epochs of fine-tuning, the gated-fusion model reaches 0.9002 accuracy, macro F1 of 0.7083 and macro AUROC of 0.9337 on the harder, multi-label PTB-XL task, with 114 of 116 body parameters transferred as described in Section III-G. In contrast, the zero-shot INCART evaluation, with no fine-tuning or alignment, collapses to macro F1 of 0.2377, a drop of 0.7087 from the 0.9465 the same model achieves on its own MIT-BIH test partition. This roughly seventy-one-point domain-shift gap is substantially larger than, though directionally consistent with, the 8.4 percent gap reported by Li, Aierken, Xu et al. for a comparable cross-database Transformer setting [26], and it exceeds both the architectural differences in Table III and the noise-induced degradation in Figure 10 by a wide margin, even allowing for the single-run caveat that applies to every number in this paper.

V. DISCUSSION

The four-axis evaluation supports three conclusions, each of which must be read in light of the beat-level split described

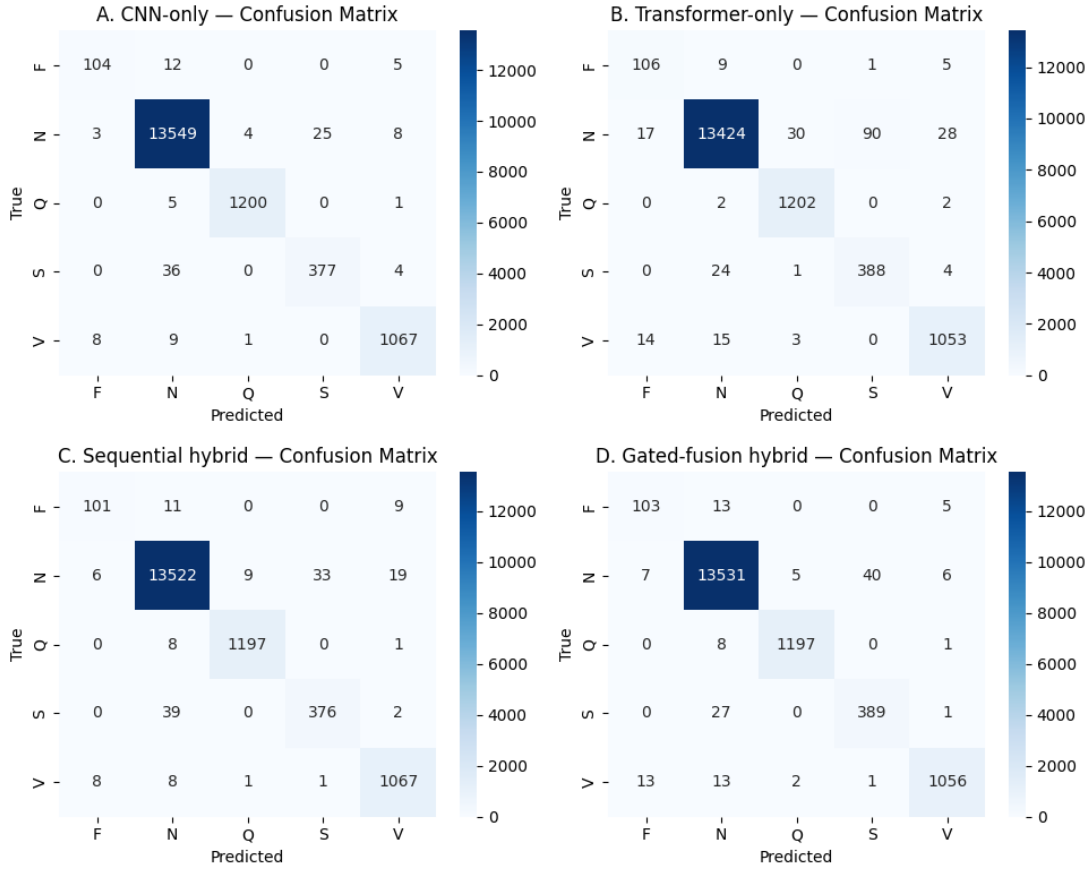


Fig. 8: Confusion matrices on the MIT-BIH test set for ablation conditions A (CNN-only, top left), B (Transformer-only, top right), C (sequential hybrid, bottom left) and D (gated-fusion hybrid, bottom right). Rows are true labels and columns are predicted labels using the AAMI class codes F, N, Q, S and V.

TABLE IV: Cross-Dataset Generalisation of the Gated-Fusion Model

Setting	Acc.	Macro-F1	Other
MIT-BIH test (source)	0.9914	0.9465	n/a
PTB-XL, aligned + fine-tuned	0.9002	0.7083	AUROC 0.9337
INCART, zero-shot	0.7329	0.2377	gap 0.7087

in Section III-D and the single-seed protocol described in Section III-E. First, additional architectural complexity does not automatically improve accuracy: the convolutional-only model achieves the highest macro F1 on this split, echoing Strodtthoff et al.’s observation that well-tuned convolutional baselines remain difficult to beat [16]. This does not make the hybrid conditions without merit, since Table III and Figure 10 together show the model ranking changes depending on whether clean accuracy, noise robustness, or cross-dataset generalisation is the criterion; a model selected purely on clean macro F1 is not necessarily best suited to a noisy or cross-institutional deployment. Because the parameter counts of the five conditions differ by close to a factor of three (Table II), part of the gap between, for example, the Transformer-only and gated-fusion conditions may also reflect capacity rather than architecture alone, and this confound is not resolved by

the present design.

Second, Figure 9 is visually consistent with Goettling et al. and Bender et al.’s conclusion that Grad-CAM alone is an insufficient basis for clinical explanation on one-dimensional ECG signals [24], [25], in that the Grad-CAM regions shown are broader and less class-specific than the Integrated Gradients or attention regions for the same beats. However, since this comparison uses a single beat per class rather than an averaged, uncertainty-quantified metric over the full test set, it should be treated as an illustration consistent with the wider literature rather than as this paper’s own statistical evidence.

Third, and most significant for clinical deployment, the zero-shot cross-dataset degradation in Table IV dwarfs both the architectural differences in Table III and the noise degradation in Figure 10. A macro F1 drop of over seventy points from MIT-BIH to INCART with no adaptation indicates domain shift, driven by lead configuration, sampling rate and population differences, as the dominant practical obstacle identified here. The partial recovery through fine-tuning and batch-normalisation alignment on PTB-XL, reaching macro AUROC 0.9337 despite a harder multi-label formulation, indicates that even a lightweight domain-alignment step [26] recovers substantial performance. This conclusion is comparatively

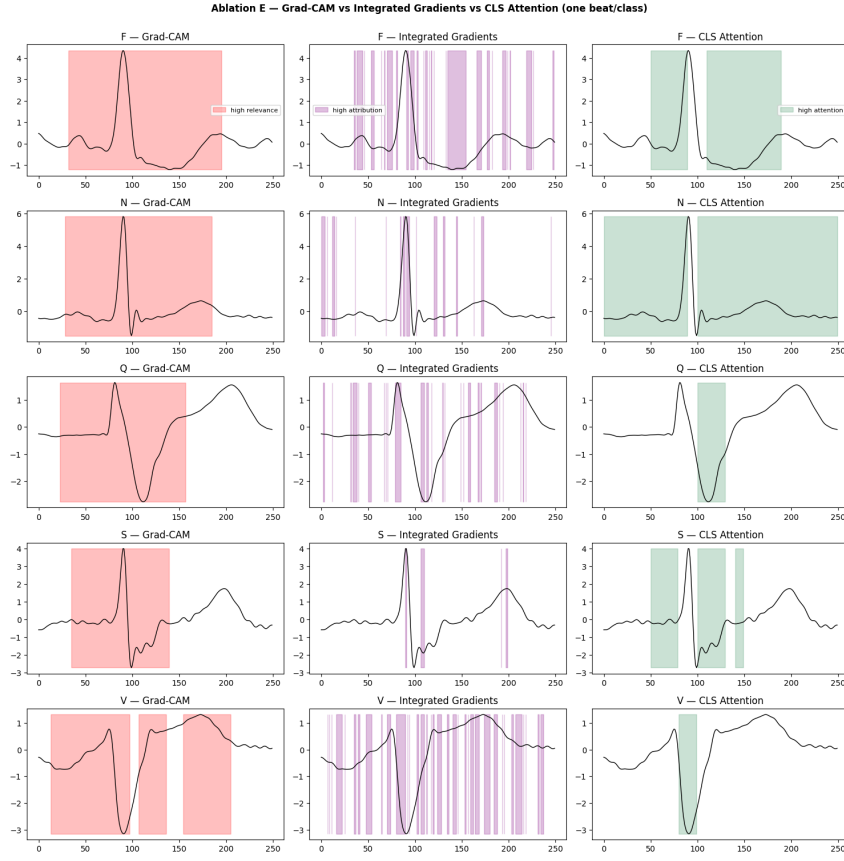


Fig. 9: Grad-CAM (left column), Integrated Gradients (middle column) and Transformer CLS attention (right column) computed on the gated-fusion model for one representative beat of each AAMI class: Fusion, Normal, Unknown/Paced, Supraventricular and Ventricular. Grad-CAM produces the broadest relevance regions; Integrated Gradients produces the sparsest. This is a single-example qualitative illustration, not an averaged, statistically validated comparison; see Section IV-B.

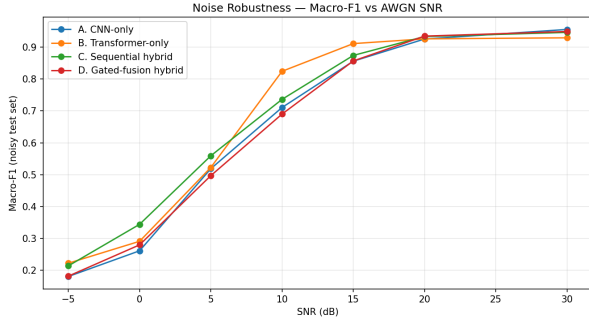


Fig. 10: Macro F1-score of the four fully supervised ablation conditions on the MIT-BIH test set under additive white Gaussian noise, swept from minus five to thirty decibels signal-to-noise ratio.

robust to the split concern raised in Section III-D, since INCART is an entirely separate database from MIT-BIH and the domain-shift gap, if anything, would only be expected to grow once the MIT-BIH-side leakage is removed and clean patient-independent test accuracy falls.

VI. LIMITATIONS AND FUTURE WORK

The methodological limitations of this study are ordered below by priority, following the assessment in Section III-D.

Patient-independence of the MIT-BIH split. The 70/15/15 split used throughout is stratified by class label at the beat level, not by patient or record, so beats from the same recording can appear in both the training and test partitions (Section III-D). This is the single most important limitation of the study: every number in Section IV that involves the MIT-BIH test partition should be treated as an optimistic upper bound until it is re-measured under a record-independent split, for example a grouped split by record identifier performed before stratification. This correction is required, not optional, before the results could support a claim of patient-level generalisation, and is the first priority for any continuation of this work.

Single-seed training. Every model in Table III was trained once with a fixed seed, so none of the reported differences come with a variance estimate, and small gaps such as the one between the gated-fusion and sequential hybrids cannot currently be distinguished from run-to-run noise. Repeating each condition across several seeds, for example five, and reporting the mean and standard deviation or a confidence

interval for macro F1, is required future work and is a comparatively inexpensive correction relative to the split issue above.

Qualitative explainability comparison. The XAI comparison in Section IV-B uses one representative beat per class and no quantitative fidelity metric, so it cannot support a claim that any one method is more faithful than another; it is presented as an illustration consistent with prior quantitative studies [24], [25], not as this paper’s own evidence. A full-test-set evaluation using perturbation-based fidelity metrics such as deletion or insertion curves, averaged with confidence intervals across all beats and classes, is required before a faithfulness ranking could be claimed.

Scale of the self-supervised baseline. Condition F is a self-supervised proxy trained from scratch at single-GPU scale rather than a fine-tuned large released checkpoint such as ECG-FM [19]; it is named a self-supervised pretrained Transformer baseline rather than a foundation model for this reason, and a genuine large-scale comparison would strengthen conclusions about pretraining value.

Scope of cross-dataset and noise testing. Cross-dataset transfer was evaluated only for the gated-fusion architecture; repeating it for all five conditions would clarify whether domain-shift robustness is itself architecture-dependent. The noise protocol injects only additive Gaussian noise; real recordings also suffer baseline wander, motion artefact and missing leads, which the TolerantECG methodology of Huynh et al. [22] suggests should also be evaluated.

VII. CONCLUSION

This paper presented a four-axis comparison of five CNN and Transformer ablation conditions for ECG arrhythmia classification: accuracy, qualitative explanation quality across three XAI methods, noise robustness, and cross-dataset generalisation through fine-tuned and zero-shot transfer. Under the beat-level MIT-BIH split used here, a convolutional-only architecture is the strongest performer on clean, in-distribution data; a learnable gated-fusion mechanism [17] is competitive with sequential hybridisation while offering a more principled feature-combination strategy, though the two are close enough that repeated-seed confirmation is needed before the difference is treated as reliable; Grad-CAM produces visibly less specific explanations than Integrated Gradients or Transformer attention on the examples examined, consistent with, though not independently confirming, prior quantitative findings; and, most importantly, uncontrolled domain shift between recording databases produces a performance collapse an order of magnitude larger than any architectural or explainability difference studied. The most consequential next step is not further architectural refinement but a record-independent re-evaluation of the MIT-BIH split, alongside repeated-seed training and a quantitative, full-test-set explainability evaluation; taken together with the present results, this would convert an informative but methodologically qualified study into one whose headline numbers can be read at face value.

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